# Essays on Heterogeneous Returns, Trust, and Household Wealth

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# Roadmap

# Roadmap

1. Introduction
2. Chapter 1
3. Chapter 2
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# Question and Unifying Theme

- This dissertation asks how heterogeneous rates of return reshape the explanation, performance, and measurement of wealth inequality.
- The common theme is that wealth inequality is not only about who saves more; it is also about who earns different returns, why, and how those differences should be evaluated.
- Chapter 1 is structural macro, Chapter 2 is empirical household finance, and Chapter 3 is inequality measurement under ambiguity.

*Core message: rates of return matter at several margins for understanding household wealth.*

# How the Three Chapters Fit Together

1. **Chapter 1**: Can return heterogeneity help a standard heterogeneous-agent model match empirical wealth moments?
2. **Chapter 2**: Is there a systematic relationship between trust and returns to net wealth, and can we still estimate a persistent household return component?
3. **Chapter 3**: If wealth is tied to social institutions and historical group structure, does the standard objective benchmark understate wealth inequality?

*Taken together, the chapters move from explaining wealth concentration, to explaining realized performance, to measuring inequality itself.*

# Roadmap

# Chapter 1: Matching Wealth Moments with Heterogeneous Returns

- I incorporate return heterogeneity into a standard heterogeneous-agent model and estimate the return distribution needed to match the wealth distribution.
- Labor income uncertainty and life-cycle structure explain part of wealth dispersion, but not enough to reproduce the empirical skewness of wealth.
- Adding return heterogeneity yields a much better fit to SCF wealth moments and implies a more realistic aggregate MPC.

Empirically, this is motivated by evidence that realized household returns vary widely across people (Fagereng et al. 2020; Daminato and Pistaferri 2024).

# Chapter 1: Main Result

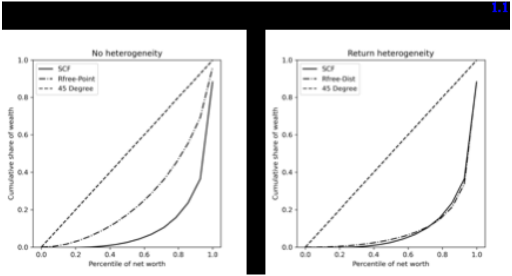



Figure: Heterogeneous returns produce a Lorenz curve much closer to the SCF target.

- Estimated uniform return distribution:

$$R = 1.0204, \qquad \nabla = 0.06833.$$

- Top 5% wealth share:
  - Data: 57.4%
  - No heterogeneity: 23.3%
  - Heterogeneity: 58.8%
- In the life-cycle model, the aggregate MPC rises substantially once return heterogeneity is introduced.

*Takeaway: return heterogeneity materially improves the fit to wealth concentration, with plausible implied dispersion in returns.*

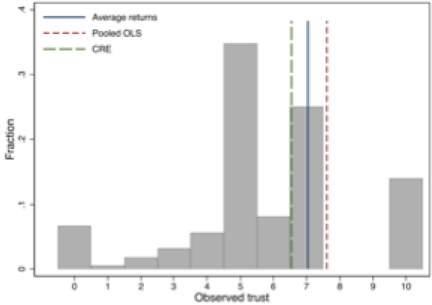
# Roadmap

# Chapter 2: Moderate Trust and Maximum Performance on Financial Investments

- I use the RAND HRS panel to construct returns to net wealth and study their relationship with generalized trust.
- Trust is measured on a 1 to 10 scale in the 2020 HRS module, while the panel structure allows estimation of a persistent household return component.
- Because trust is time-invariant in the HRS, the correlated random effects (CRE) framework is especially useful here.

The broader empirical motivation is that trust is often positively related to economic performance, but the relationship need not be monotone.

# Chapter 2: Main Result




[redacted]

Figure: Estimated return-maximizing trust levels lie in a moderate region of the observed trust distribution.

| Specification | Avg. returns | Pooled OLS | CRE |
|---|---|---|---|
| Trust* without controls | 6.70 | 6.89 | 6.34 |
| Trust* with controls | 7.07 | 7.61 | 6.59 |

- The fitted relationship between trust and returns to net wealth is hump-shaped.
- The maximum is consistently in a moderate range, roughly between 6 and 7 in the preferred specifications.
- The persistent component remains economically important, and portfolio-share controls change the estimated fixed-effect distribution only modestly.

# Roadmap

# Chapter 3: Wealth Inequality Under Ambiguity

- Motivated by Rawls' discussion of justice behind a veil of ignorance, I propose a wealth inequality measure based on choice under ambiguity.
- The key idea is that wealth distributions may need to be ranked using socially relevant states, rather than objective uncertainty alone.
- I then adapt the social-welfare approach to inequality measurement to define an ambiguity-based wealth inequality measure.

$$I^A = 1 - \frac{w^{EDE}}{\mu}$$

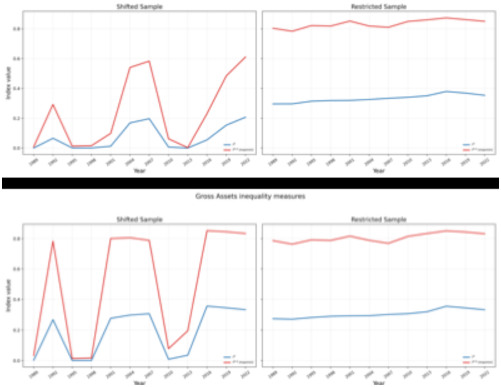
*Question: does the standard objective benchmark understate wealth inequality when racial structure is central?*

# Chapter 3: Data Pattern

[redacted]


- The raw data show a persistent and economically large racial wealth gap in both net wealth and gross assets.
- This is the empirical benchmark the theoretical inequality measures should speak to.

# Chapter 3: Ambiguity-Based Wealth Inequality




- The objective benchmark understates wealth inequality relative to the ambiguity-based measure for the same wealth distribution.
- This finding holds for both net wealth and gross assets, and the measures track the broad time pattern of the median wealth-gap diagnostics.

# Roadmap

# Conclusion

- **Chapter 1**: return heterogeneity can match key wealth moments in a structural macro model, with an estimated distribution close to empirical evidence.
- **Chapter 2**: returns to net wealth exhibit a hump-shaped relationship with trust, and the maximizing level of trust lies in a moderate range between 6 and 7.
- **Chapter 3**: the ambiguity-based wealth inequality measure implies more inequality than the standard objective benchmark for the same wealth distribution.

*Overall conclusion: heterogeneous returns are central both to explaining household wealth inequality and to evaluating it.*

# Roadmap

# Backup: Chapter 2 Interpretation

- The Chapter 2 result should not be read as “more trust is always better”.
- Instead, the evidence points to a moderate amount of trust being associated with the highest fitted returns to net wealth.
- The preferred CRE specification keeps that relationship while also accommodating persistent cross-household return heterogeneity.

Back to Chapter 2

# Backup: Chapter 3 Interpretation

- The ambiguity-based ranking is useful when the social planner cares about worst-performing group outcomes under uncertainty about relevant states.
- In the data, this shifts measured inequality upward relative to the objective benchmark.
- So the contribution is not only a new formula; it is a different normative benchmark for wealth inequality.

Back to Chapter 3